GRAVITY AND AIR RESISTANCE

LEARNING GOALS

Students will:

- Solve problems where the acceleration is constant
- Use the acceleration due to gravity (9.81 m/s^2) to solve problems.
- Describe how air resistance can change how objects accelerate as they fall.
- Explain when it is reasonable to ignore air resistance in problem solving.

CONSTANT ACCELERATION

Many problems in physics have a constant acceleration, or can be divided into parts that have a constant acceleration. When the acceleration is constant, we have a number of other equations that help us solve problems. REMEMBER, THE EQUATIONS THAT FOLLOW ARE ONLY VALID WHEN THE ACCELERATION IS **CONSTANT.**

$$\vec{v_2} = \vec{v_1} + \vec{a}\,\Delta t$$

$$\overrightarrow{v_{ave}} = \frac{\vec{v_2} + \vec{v_1}}{2}$$

$$\overrightarrow{\Delta d} = \frac{\vec{v_2} + \vec{v_1}}{2}\Delta t$$

$$\overrightarrow{\Delta d} = \vec{v_1}\Delta t + \frac{1}{2}\vec{a}\Delta t^2$$

$$\overrightarrow{\Delta d} = \vec{v_2}\Delta t - \frac{1}{2}\vec{a}\Delta t^2$$

$$\vec{v_2}^2 = \vec{v_1}^2 + 2\vec{a}\,\overrightarrow{\Delta d}$$

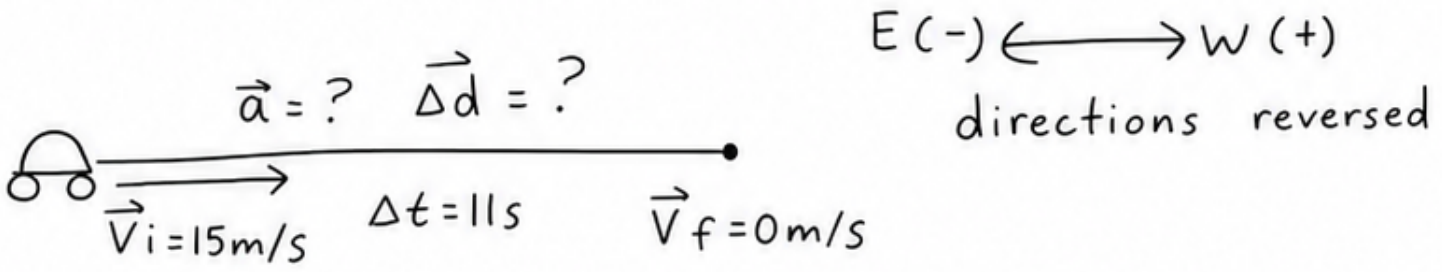
EXAMPLE

A car is approaching an intersection at 15 m/s [W]. It brakes and comes to a stop in 11s.

a) What was the car's acceleration?

b) What was the car's stopping distance?

Draw a sketch showing the car, indicating that west is positive, the initial velocity, the time, the final velocity and all variables you need to find:



Solution: a) $\overrightarrow{a_{ave}} = \frac{\vec{v_2} - \vec{v_1}}{\Delta t} = \frac{0 - 15\frac{m}{s}}{11s} = -1.36\frac{m}{s^2}$ [W] Therefore the acceleration is 1.4 m/s^2 [E].

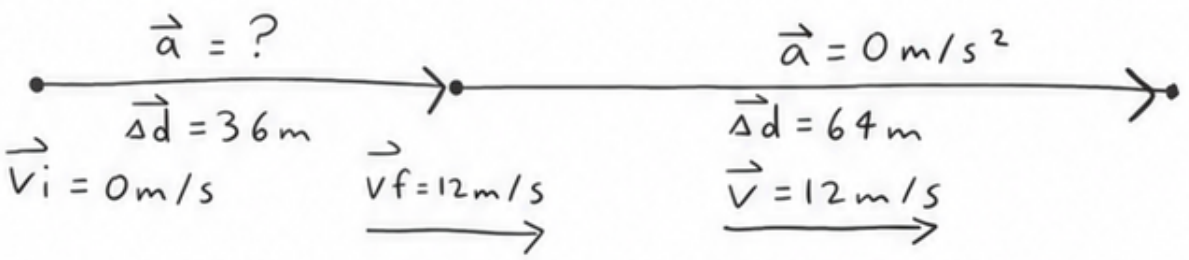
b) $\overrightarrow{\Delta d} = \frac{\vec{v_2} + \vec{v_1}}{2}\Delta t = \frac{0 + 15\frac{m}{s}}{2}(11s) = 82.5\,m$ [W] Therefore the car stops in 83 m.

EXAMPLE

↳ exact value

A sprinter starts from rest in a 100 m race. She runs the first 36 m of the race at a constant acceleration reaching a speed of 12 m/s. She holds this speed until she finishes the race. What is her total time?

Sketch a diagram of the race, clearly indicating the two parts and labelling all the quantities you can. 1. 2 W- ⟷ E+



Solution: This race does not have a constant acceleration throughout, but it can be divided into two parts each of which does have a constant acceleration:

Find time for the first part of the race:

$$\Delta t = \frac{\vec{\Delta d}}{\vec{v}_{avg}} \qquad \Delta t_1 = 6\ s$$

$$\vec{V}_{avg} = \frac{\vec{V}_f + \vec{v}_i}{2}$$

$$\vec{V}_{avg} = \frac{12}{2}$$

$$\vec{V}_{avg} = 6\ m/s\ [E]$$

$$\Delta t = \frac{36}{6}$$

Now for the second part of the race:

$$\Delta t = \frac{\vec{\Delta d}}{\vec{v}} \qquad \Delta t = \frac{64}{12}$$

$$\Delta t_2 = 5.3\ s$$

Therefore her time for the race is...

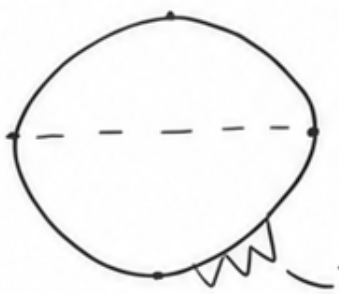
$$\Delta t_{total} = \Delta t_1 + \Delta t_2 \qquad \Delta t_{total} = 11.3\ s$$

$\Delta t_1 = 6.0$ s $\Delta t_2 = 5.3$ s $\Delta t_{tot} = 11.3$ s

THE ACCELERATION DUE TO GRAVITY

1. On or near the earth's surface, the acceleration due to gravity is essentially a constant value of 9.8 m/s². It varies by a small amount depending on where you are on the earth. **Explain why.**

- Distance from Earth's centre $\propto \frac{1}{\sqrt{g}}$



→ earth is oblong so distance from the centre varies

→ Also, mountains affect the distance from the earth's centre

SOLVE THIS PROBLEM

2. A ball is thrown up into the air with an initial velocity of 16 m/s [up].

a) Sketch a diagram of the situation, labelling all known quantities. Clearly indicate which way is positive.

b) What is the velocity of the ball after

i) 0.95 s?

ii) 2.3 s?

c) What is the maximum height reached by the ball?

$\vec{V}_2 = \vec{V}_1 + \vec{a}\,\Delta t$

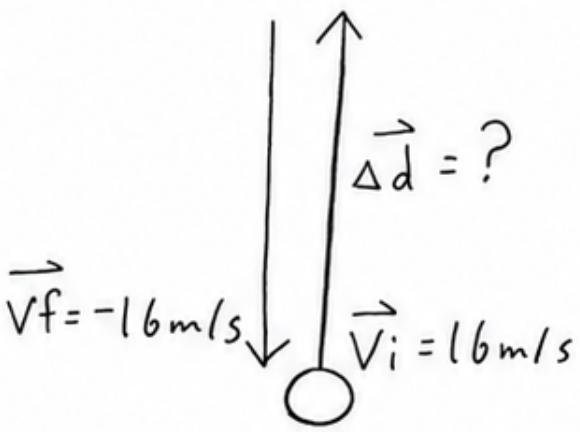
$\vec{V}_2 = 16 + -9.8 \times 0.95$

$V_2 = +6,7 \text{ m/s at } 0.95 \text{ s}$

$\vec{V}_2 = \vec{V}_1 + \vec{a}\,\Delta t$

$\vec{V}_2 = 16 + (-9.8)(2.3)$

$\vec{V}_2 = -6.5 \text{ m/s at } 2.3 \text{ s}$



$\vec{a} = -9.8 \text{ m/s}^2$

$\vec{V}_2^{\,2} = \vec{V}_1^{\,2} + 2\vec{a}\,\Delta\vec{d}$

$$\frac{\vec{V}_2^{\,2} - \vec{V}_i^{\,2}}{2a} = \Delta\vec{d}$$

$\Delta d = +13 \text{ m}$

b) i)6.7 m/s [up] ii)6.5 m/s [down] (2 sig Dig) c) 13 m

AIR RESISTANCE

3. In the problem you solved above, you ignored air resistance. Consider what would change in your answers if you took air resistance into account. How would your values for velocity and maximum height change if we were to include air resistance? Why?

Air resistance adds a force that opposes the ball's motion, so the velocities would change (not necessarily to smaller magnitudes) and the maximum height would also be lower.

Air resistance becomes greater as you speed up. The force of air resistance on an object is proportional to the square of the objects speed. The faster you go, the more air resistance there is.

$$F_{air} \propto v^2$$

4. Based on this relationship, why would it be impossible for us to take air resistance into account when doing calculations with the equations given on page one?

($F_{drag} = \frac{1}{2} \rho A C (v^2)$) It would be impossible to take drag into account because it depends on velocity, and the velocity changes, so the force of drag would be constantly changing, which would cause the acceleration to not be constant,

5. What is the term we use to describe an object's motion when the force of air resistance is EQUAL to the force of gravity? (Hint: consider what would happen to the object's speed at this point) The term is called terminal speed

$F_{net} = mg - F_{drag} \rightarrow F_{net} = mg - mg$

$F_{drag} = mg$ ↗ plug in $F_{net} = 0$

our equations require a constant acceleration

$F_{net} = ma$

$\frac{F_{net}}{m} = a$ $\frac{0}{m} = a$ $\therefore a = 0$, so speed stays constant

HOMEWORK

New Text: p19 #1,3,4, page 20 # 1-5, p 21#3,4, 7